Equal real Eigenvalues with multiplicity two

Previously we looked at how to solve systems when the characteristic equation produced distinct real Eigenvalues. Now we want to turn our attention toward equal real Eigenvalues.

For equal real Eigenvalues, there are two cases we want to look at.

1. The repeated Eigenvalue produces multiple linearly independent Eigenvectors

2. The repeated Eigenvalue produces only one Eigenvector

In the first case, the solution to the system will be just like the solution for distinct real Eigenvalues, $\overrightarrow{x} = c_1\overrightarrow{x_1} + c_2\overrightarrow{x_2} + \ldots + c_n\overrightarrow{x_n}$. But in the second case, we'll still use the single Eigenvector for the solution $\overrightarrow{x_1}$, but then we'll need a special equation,

$$(A - \lambda_1 I)\overrightarrow{p_1} = \overrightarrow{k_1}$$

to find the second solution

$$\overrightarrow{x_2} = \overrightarrow{k_1}te^{\lambda_1 t} + \overrightarrow{p_1}e^{\lambda_1 t}$$

Substituting the single Eigenvector into $(A - \lambda_1 I)\overrightarrow{p_1} = \overrightarrow{k_1}$ for $\overrightarrow{k_1}$ allows us to find a second vector $\overrightarrow{p_1}$ that we can use for the second solution, $\overrightarrow{x_2}$.

Let's start by looking at these two cases for an Eigenvalue with multiplicity two.

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Eigenvalue with multiplicity two

An **Eigenvalue with multiplicity two** is an Eigenvalue that shows up twice in the characteristic equation.

Let's do an example with the first case above, where a repeated Eigenvalue produces multiple linearly independent Eigenvectors.

Example

Find the general solution to the system.

$$\vec{x}' = \begin{bmatrix} 3 & -1 & -1 \\ 1 & 1 & -1 \\ 1 & -1 & 1 \end{bmatrix} \vec{x}$$

We'll need to start by finding the matrix $A - \lambda I$,

$$A - \lambda I = \begin{bmatrix} 3 & -1 & -1 \\ 1 & 1 & -1 \\ 1 & -1 & 1 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 3 & -1 & -1 \\ 1 & 1 & -1 \\ 1 & -1 & 1 \end{bmatrix} - \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 3-\lambda & -1 & -1 \\ 1 & 1-\lambda & -1 \\ 1 & -1 & 1-\lambda \end{bmatrix}$$

and then find its determinant $|A - \lambda I|$.

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$$|A-\lambda I| = \begin{vmatrix} 3-\lambda & -1 & -1 \\ 1 & 1-\lambda & -1 \\ 1 & -1 & 1-\lambda \end{vmatrix}$$

$$|A-\lambda I| = (3-\lambda)\begin{vmatrix} 1-\lambda & -1 \\ -1 & 1-\lambda \end{vmatrix} - (-1)\begin{vmatrix} 1 & -1 \\ 1 & 1-\lambda \end{vmatrix} + (-1)\begin{vmatrix} 1 & 1-\lambda \\ 1 & -1 \end{vmatrix}$$

$$|A-\lambda I| = (3-\lambda)[(1-\lambda)(1-\lambda)-(-1)(-1)]$$

$$+[(1)(1-\lambda)-(-1)(1)]-[(1)(-1)-(1-\lambda)(1)]$$

$$|A-\lambda I| = (3-\lambda)(1-2\lambda+\lambda^2-1)+(1-\lambda+1)-(-1-1+\lambda)$$

$$|A-\lambda I| = (3-\lambda)(-2\lambda+\lambda^2)+1-\lambda+1+1+1-\lambda$$

$$|A-\lambda I| = -6\lambda+3\lambda^2+2\lambda^2-\lambda^3+4-2\lambda$$

$$|A-\lambda I| = 4-8\lambda+5\lambda^2-\lambda^3$$

Solve the characteristic equation for the Eigenvalues.

$$4-8\lambda+5\lambda^2-\lambda^3=0$$

$$\lambda^3-5\lambda^2+8\lambda-4=0$$

$$(\lambda-1)(\lambda-2)(\lambda-2)=0$$

$$\lambda = 1,\ 2,\ 2$$

We'll handle $\lambda_1 = 1$ first, starting with finding $A-1I$.

$$A-1I = \begin{bmatrix} 3-1 & -1 & -1 \\ 1 & 1-1 & -1 \\ 1 & -1 & 1-1 \end{bmatrix}$$

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$$A - 1I = \begin{bmatrix} 2 & -1 & -1 \\ 1 & 0 & -1 \\ 1 & -1 & 0 \end{bmatrix}$$

Put this matrix into reduced row-echelon form.

$$\begin{bmatrix} 1 & 0 & -1 \\ 2 & -1 & -1 \\ 1 & -1 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & -1 \\ 0 & -1 & 1 \\ 1 & -1 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & -1 \\ 0 & -1 & 1 \\ 0 & -1 & 1 \end{bmatrix} \rightarrow$$

$$\begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & -1 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

If we turn this matrix into a system of equations, we get

$$k_1 - k_3 = 0$$

$$k_2 - k_3 = 0$$

or

$$k_1 = k_3$$

$$k_2 = k_3$$

So if we choose $k_3 = 1$, we get $k_1 = 1$ and $k_2 = 1$, or

$$\overrightarrow{k_1} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

and therefore

$$\overrightarrow{x_1} = \overrightarrow{k_1} e^{\lambda_1 t}$$

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$$\overrightarrow{x_1} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} e^t$$

Then for the repeated Eigenvalues $\lambda_2 = \lambda_3 = 2$, we find

$$A - 2I = \begin{bmatrix} 3-2 & -1 & -1 \\ 1 & 1-2 & -1 \\ 1 & -1 & 1-2 \end{bmatrix}$$

$$A - 2I = \begin{bmatrix} 1 & -1 & -1 \\ 1 & -1 & -1 \\ 1 & -1 & -1 \end{bmatrix}$$

Put this matrix into reduced row-echelon form.

$$\begin{bmatrix} 1 & -1 & -1 \\ 0 & 0 & 0 \\ 1 & -1 & -1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -1 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

If we turn this first row into an equation, we get

$$k_1 - k_2 - k_3 = 0$$

$$k_1 = k_2 + k_3$$

We'll choose $(k_2, k_3) = (1,0)$, which results in $k_1 = 1$.

$$\overrightarrow{k_2} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

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and therefore

$$\overrightarrow{x_2} = \overrightarrow{k_2} e^{\lambda_2 t}$$

$$\overrightarrow{x_2} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} e^{2t}$$

But we can actually find a second linearly independent Eigenvector from $k_1 = k_2 + k_3$. If we choose $(k_2, k_3) = (0,1)$ instead, we get $k_1 = 1$, and we find a second Eigenvector.

$$\overrightarrow{k_3} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

and therefore

$$\overrightarrow{x_3} = \overrightarrow{k_3} e^{\lambda_3 t}$$

$$\overrightarrow{x_3} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} e^{2t}$$

These two Eigenvectors $\overrightarrow{k_2} = (1,1,0)$ and $\overrightarrow{k_3} = (1,0,1)$ are linearly independent of one another, because there's no constant by which we can multiply either vector that will give us the other vector as a result. Therefore, the general solution will be

$$\overrightarrow{x} = c_1\overrightarrow{x_1} + c_2\overrightarrow{x_2} + c_3\overrightarrow{x_3}$$

$$\overrightarrow{x} = c_1 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} e^{t} + c_2 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} e^{2t} + c_3 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} e^{2t}$$

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Now let's do an example with equal real Eigenvalues, but this time where we find only one Eigenvector associated with the two equal Eigenvalues.

Example

Find the general solution to the system of differential equations.

$$x_1'(t) = 2x_1 - x_2$$

$$x_2'(t) = x_1 + 4x_2$$

We can represent this system in matrix form as

$$\vec{x}' = A\vec{x}$$

$$\vec{x}' = \begin{bmatrix} 2 & -1 \\ 1 & 4 \end{bmatrix} \vec{x}$$

We'll need to find the matrix $A - \lambda I$,

$$A - \lambda I = \begin{bmatrix} 2 & -1 \\ 1 & 4 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 2 & -1 \\ 1 & 4 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 2-\lambda & -1 \\ 1 & 4-\lambda \end{bmatrix}$$

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and then find its determinant $|A - \lambda I|$.

$$|A - \lambda I| = \begin{vmatrix} 2-\lambda & -1 \\ 1 & 4-\lambda \end{vmatrix}$$

$$|A - \lambda I| = (2-\lambda)(4-\lambda) - (-1)(1)$$

$$|A - \lambda I| = 8 - 2\lambda - 4\lambda + \lambda^2 + 1$$

$$|A - \lambda I| = \lambda^2 - 6\lambda + 9$$

Solve the characteristic equation for the Eigenvalues.

$$\lambda^2 - 6\lambda + 9 = 0$$

$$(\lambda - 3)(\lambda - 3) = 0$$

$$\lambda = 3,\ 3$$

Then for these Eigenvalues, $\lambda_1 = \lambda_2 = 3$, we find

$$A - 3I = \begin{bmatrix} 2-3 & -1 \\ 1 & 4-3 \end{bmatrix}$$

$$A - 3I = \begin{bmatrix} -1 & -1 \\ 1 & 1 \end{bmatrix}$$

Put this matrix into reduced row-echelon form.

$$\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$$

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If we turn this first row into an equation, we get

$$k_1 + k_2 = 0$$

$$k_1 = -k_2$$

We'll choose $k_2 = -1$, which results in $k_1 = 1$.

$$\overrightarrow{k_1} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Other than the trivial solution $\overrightarrow{k_1} = (0,0)$, any other (k_1, k_2) pair we find from $k_1 = -k_2$ will result in a vector that's linearly dependent with the $\overrightarrow{k_1} = (1, -1)$ vector we already found. So we can say that the Eigenvalue $\lambda_1 = \lambda_2 = 3$ produces only one Eigenvector.

$$\overrightarrow{x_1} = \overrightarrow{k_1} e^{\lambda_1 t}$$

$$\overrightarrow{x_1} = \begin{bmatrix} 1 \\ -1 \end{bmatrix} e^{3t}$$

Because we only find one Eigenvector for the two Eigenvalues $\lambda_1 = \lambda_2 = 3$, we have to use $\overrightarrow{k_1} = (1, -1)$ to find a second solution.

$$(A - \lambda_1 I)\overrightarrow{p_1} = \overrightarrow{k_1}$$

$$\begin{bmatrix} -1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} p_1 \\ p_2 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Expanding this matrix equation as a system of equations gives

$$-p_1 - p_2 = 1$$

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$$p_1 + p_2 = -1$$

Because this system is equivalent to just one equation (because multiplying through the second equation by -1 results in the first equation), there are an infinite number of value pairs we can choose for $\overrightarrow{p_1} = (p_1, p_2)$. Let's pick simple values, like $\overrightarrow{p_1} = (1, -2)$.

Then our second solution will be

$$\overrightarrow{x_2} = \overrightarrow{k_1} te^{\lambda_1 t} + \overrightarrow{p_1} e^{\lambda_1 t}$$

$$\overrightarrow{x_2} = \begin{bmatrix} 1 \\ -1 \end{bmatrix} te^{3t} + \begin{bmatrix} 1 \\ -2 \end{bmatrix} e^{3t}$$

and the general solution to the homogeneous system is

$$\overrightarrow{x} = c_1 \overrightarrow{x_1} + c_2 \overrightarrow{x_2}$$

$$\overrightarrow{x} = c_1 \begin{bmatrix} 1 \\ -1 \end{bmatrix} e^{3t} + c_2 \left(\begin{bmatrix} 1 \\ -1 \end{bmatrix} te^{3t} + \begin{bmatrix} 1 \\ -2 \end{bmatrix} e^{3t} \right)$$

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